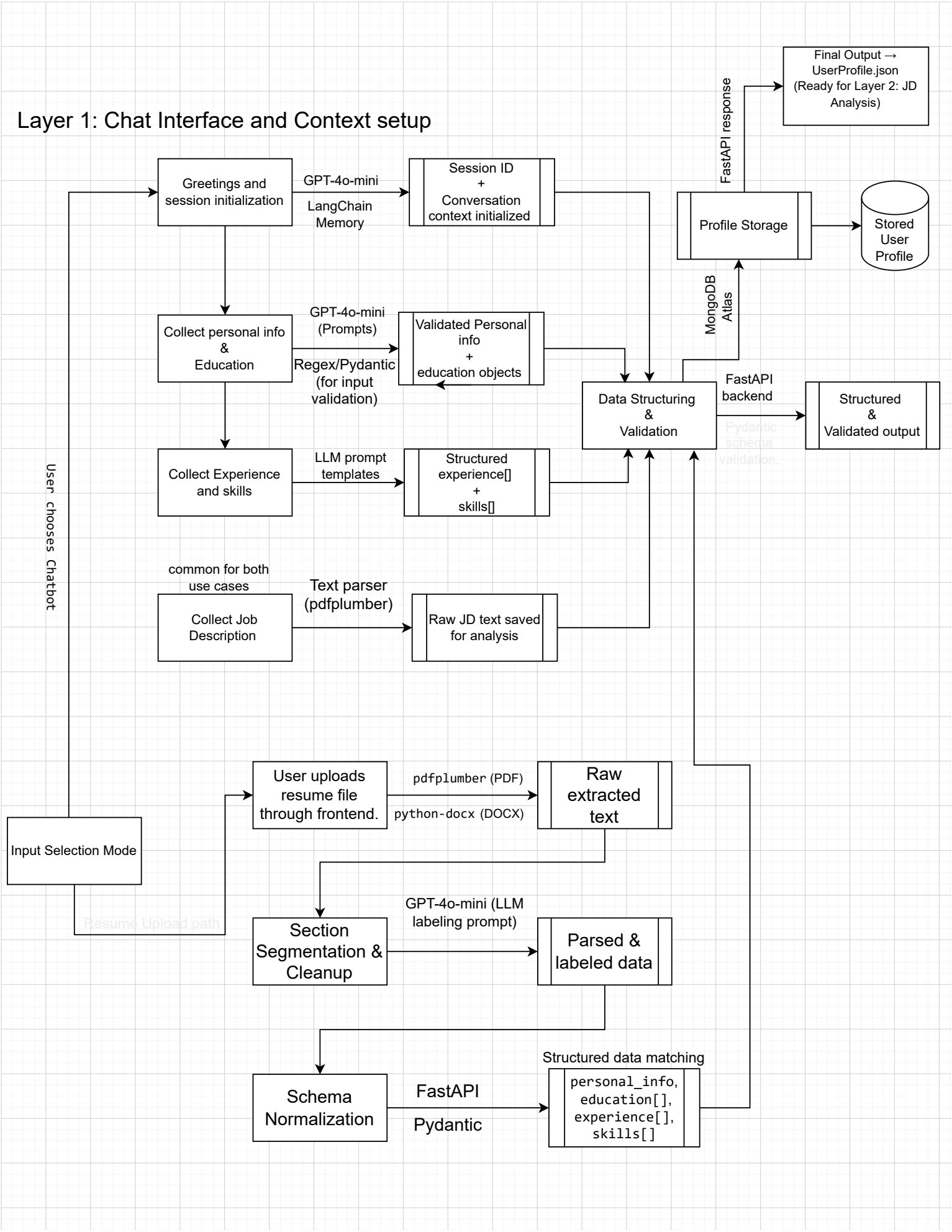
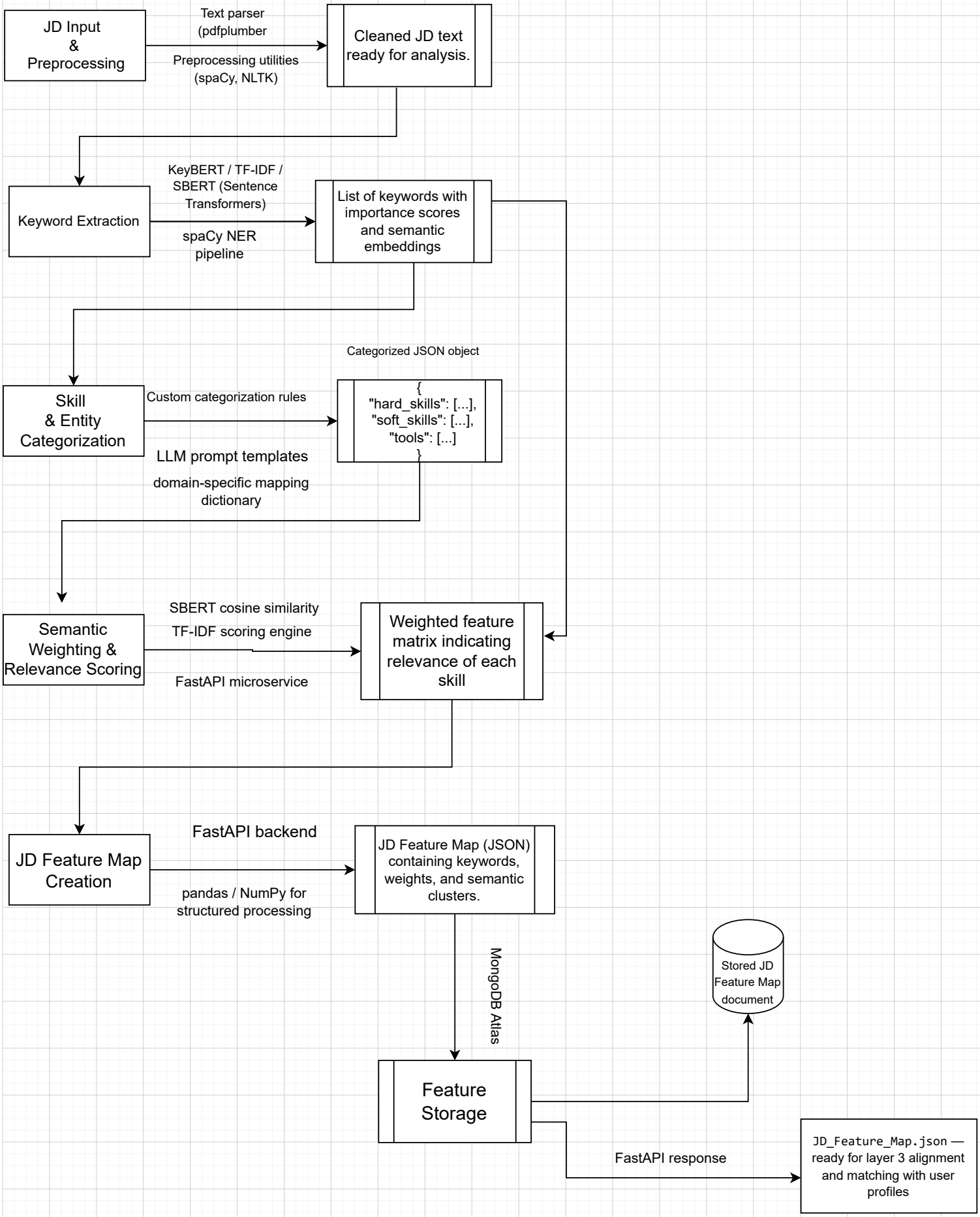
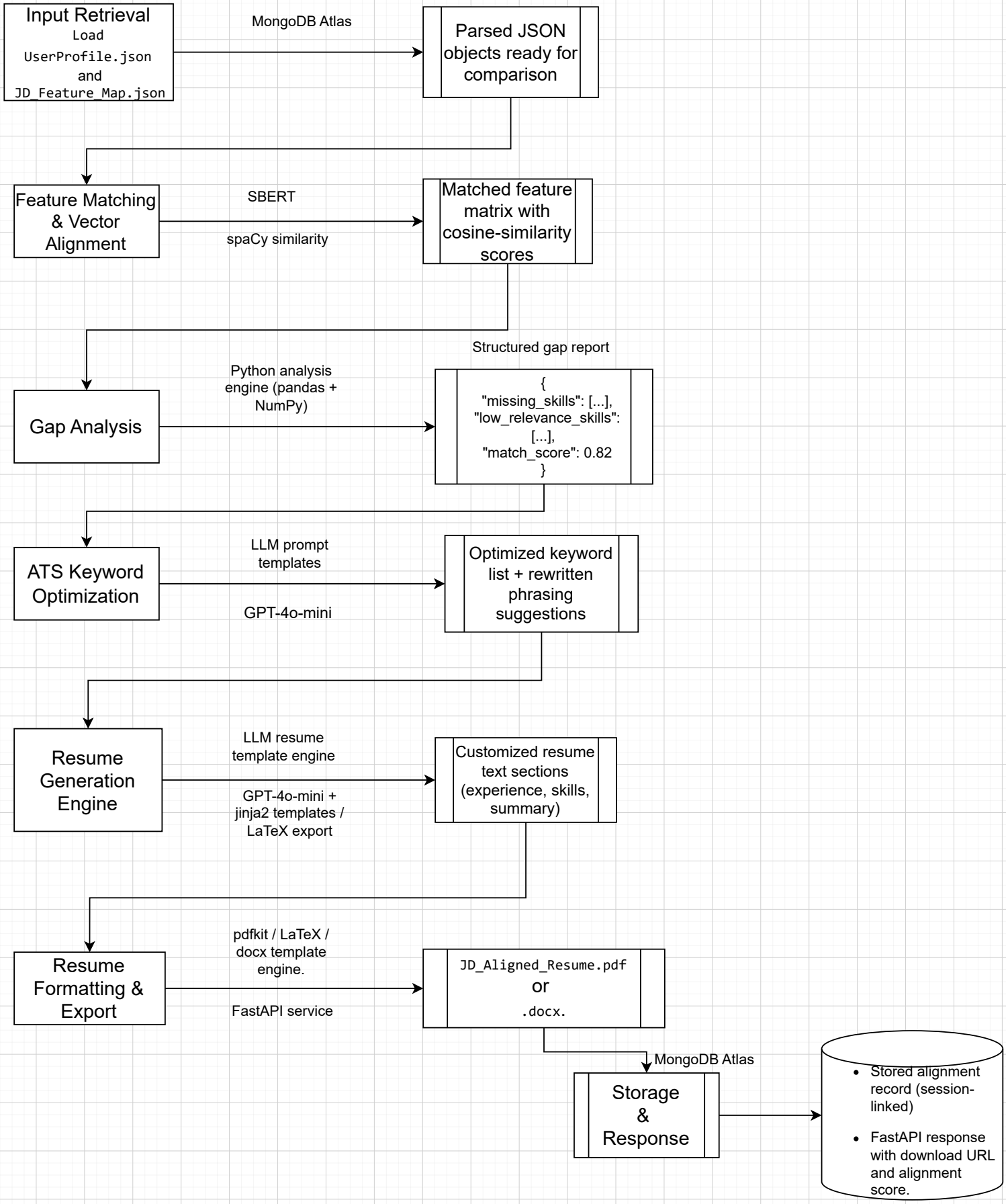
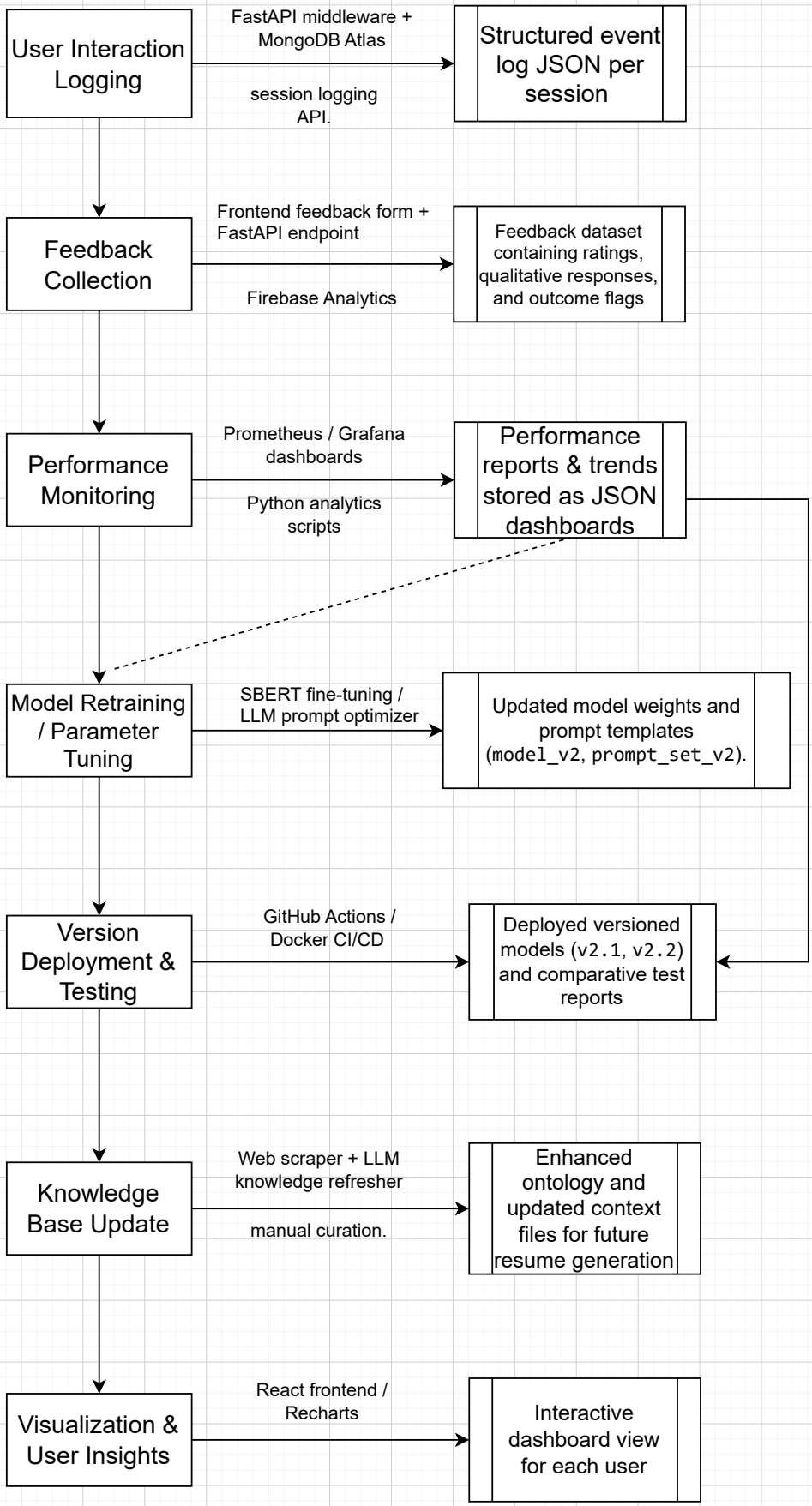

Layer 1: Chat Interface and Context setup
Greetings and session initialization
GPT-4o-mini
LangChain Memory
Session ID + Conversation context initialized
Collect personal info & Education
GPT-4o-mini (Prompts)
Regex/Pydantic (for input validation)
Validated Personal info + education objects
Collect Experience and skills
LLM prompt templates
Structured experience[] + skills[]
common for both use cases
Collect Job Description
Text parser (pdfplumber)
Raw JD text saved for analysis
Data Structuring & Validation
FastAPI backend
Pydantic schema validation.
Structured & Validated output
MongoDB Atlas
Profile Storage
FastAPI response
Final Output → UserProfile.json (Ready for Layer 2: JD Analysis)
Stored User Profile
User chooses Chatbot
Input Selection Mode
Resume Upload path
User uploads resume file through frontend.
pdfplumber (PDF)
python-docx (DOCX)
Raw extracted text
Section Segmentation & Cleanup
GPT-4o-mini (LLM labeling prompt)
Parsed & labeled data
Schema Normalization
FastAPI
Pydantic
Structured data matching
personal_info, education[], experience[], skills[]

Layer 2: JD Intelligence & Keyword Extraction



Layer 3: Alignment Engine & Resume Generation



Layer 4: User Feedback & Continuous Optimization